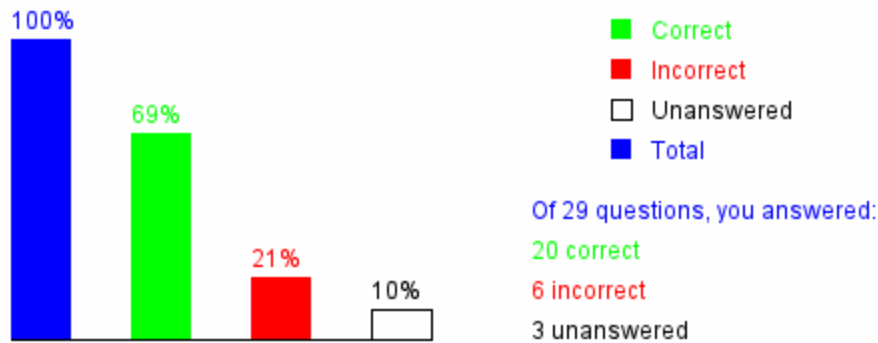


Introduction to Programming Using Python, Y. Daniel Liang

Many questions in this edition have been updated in the new edition. Please check with the publisher on the newest edition.

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Chapter 5 Loops



Please send suggestions and errata to Dr. Liang at y.daniel.liang@gmail.com. Indicate which book and edition you are using. Thanks!

Section 5.2 The while Loop

5.1 How many times will the following code print "Welcome to Python"?

```
count = 0
while count < 10:
    print("Welcome to Python")
    count += 1
```

- ☐ A. 8
- ☐ B. 9
- ☒ C. 10
- ☐ D. 11
- ☐ E. 0

Your answer is correct



5.2 What is the output of the following code?

```
x = 0
while x < 4:
    x = x + 1

print("x is", x)
```

- ☐ A. x is 0
- ☐ B. x is 1
- ☐ C. x is 2
- ☐ D. x is 3
- ☒ E. x is 4

Your answer is correct



5.3 Analyze the following code.

```
count = 0
while count < 100:
    # Point A
    print("Welcome to Python!")
    count += 1
    # Point B

# Point C
```

- ☒ A. count < 100 is always True at Point A
☐ B. count < 100 is always True at Point B
☒ C. count < 100 is always False at Point B
☐ D. count < 100 is always True at Point C
☒ E. count < 100 is always False at Point C

Your answer ACE is incorrect 
[Click here to show the correct answer](#)

5.4 How many times will the following code print "Welcome to Python"?

```
count = 0
while count < 10:
    print("Welcome to Python")
```

- ☐ A. 8
☐ B. 9
☐ C. 10
☐ D. 11
☒ E. infinite number of times

Your answer is correct 

5.5 What will be displayed when the following code is executed?

```
number = 6
while number > 0:
    number -= 3
    print(number, end = ' ')
```

- ☐ A. 6 3 0
☐ B. 6 3
☒ C. 3 0
☐ D. 3 0 -3
☐ E. 0 -3

Your answer is correct 

Section 5.3 The for Loop

5.6 Analyze the following statement:

```
sum = 0
for d in range(0, 10, 0.1):
    sum += sum + d
```

- ☐ A. The program has a syntax error because the range function cannot have three arguments.

- ☒ B. The program has a syntax error because the arguments in the range must be integers.
- ☐ C. The program runs in an infinite loop.
- ☐ D. The program runs fine.

Your answer is correct



5.7 Which of the following loops prints "Welcome to Python" 10 times?

A:
`for count in range(1, 10):`
`print("Welcome to Python")`

B:
`for count in range(0, 10):`
`print("Welcome to Python")`

C:
`for count in range(1, 11):`
`print("Welcome to Python")`

D:
`for count in range(1, 12):`
`print("Welcome to Python")`

- ☐ A. BD
- ☐ B. ABC
- ☐ C. AC
- ☒ D. BC
- ☐ E. AB

Your answer is correct



5.8 The function `range(5)` return a sequence _____.

- ☐ A. 1, 2, 3, 4, 5
- ☐ B. 0, 1, 2, 3, 4, 5
- ☐ C. 1, 2, 3, 4
- ☒ D. 0, 1, 2, 3, 4

Your answer is correct



5.9 Which of the following function returns a sequence 0, 1, 2, 3?

- ☐ A. `range(0, 3)`
- ☒ B. `range(0, 4)`
- ☐ C. `range(3)`
- ☒ D. `range(4)`

Your answer is correct



5.10 Which of the following function is incorrect?

- ☒ A. `range(0, 3.5)`
- ☐ B. `range(10, 4, -1)`
- ☐ C. `range(1, 3, 1)`
- ☒ D. `range(2.5, 4.5)`

☒ E. `range(1, 2.5, 4.5)`

Your answer is correct



5.11 Which of the following loops correctly computes $1/2 + 2/3 + 3/4 + \dots + 99/100$?

A:

```
sum = 0
for i in range(1, 99):
    sum += i / (i + 1)
```

```
print("Sum is", sum)
```

B:

```
sum = 0
for i in range(1, 100):
    sum += i / (i + 1)
```

```
print("Sum is", sum)
```

C:

```
sum = 0
for i in range(1.0, 99.0):
    sum += i / (i + 1)
```

```
print("Sum is", sum)
```

D:

```
sum = 0
for i in range(1.0, 100.0):
    sum += i / (i + 1)
```

```
print("Sum is", sum)
```

- ☐ A. BCD
- ☐ B. ABCD
- ☒ C. B
- ☐ D. CDE
- ☐ E. CD

Your answer is correct



5.12 The following loop displays _____.

```
for i in range(1, 11):
    print(i, end = " ")
```

- ☐ A. 1 2 3 4 5 6 7 8 9
- ☒ B. 1 2 3 4 5 6 7 8 9 10
- ☐ C. 1 2 3 4 5
- ☐ D. 1 3 5 7 9
- ☐ E. 2 4 6 8 10

Your answer is correct



5.13 What is the output for y?

```
y = 0
for i in range(0, 10):
    y += i
```

print(y)

- ☐ A. 10
- ☐ B. 11
- ☐ C. 12
- ☐ D. 13
- ☒ E. 45

Your answer is correct



[Click here to show an explanation](#)

5.14 What is the output for y?

```
y = 0
for i in range(0, 10, 2):
    y += i
```

print(y)

- ☒ A. 9
- ☐ B. 10
- ☐ C. 11
- ☐ D. 20

Your answer A is incorrect



[Click here to show the correct answer](#)

5.15 What is the output for y?

```
y = 0
for i in range(10, 1, -2):
    y += i
```

print(y)

- ☐ A. 10
- ☐ B. 40
- ☒ C. 30
- ☐ D. 20

Your answer is correct



5.16 Given the following four patterns,

Pattern A

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4 5 6
```

Pattern B

```
1 2 3 4 5 6
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

Pattern C

```
1
2 1
3 2 1
4 3 2 1
5 4 3 2 1
6 5 4 3 2 1
```

Pattern D

```
1 2 3 4 5 6
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

Which of the pattern **is** produced by the following code?

```
for i in range(1, 6 + 1):
    for j in range(6, 0, -1):
```

```
print(j if j <= i else " ", end = " ")
print()
```

- ☐ A. Pattern A
- ☐ B. Pattern B
- ☒ C. Pattern C
- ☐ D. Pattern D

Your answer is correct



Section 5.5 Minimizing Numerical Errors

5.17 Analyze the following fragment:

```
sum = d = 0
while d != 10.0:
    d += 0.1
    sum += sum + d
```

- ☐ A. The program does not run because sum and d are not initialized correctly.
- ☐ B. The program never stops because d is always 0.1 inside the loop.
- ☒ C. The program may not stop because of the phenomenon referred to as numerical inaccuracy for operating with floating-point numbers.
- ☐ D. After the loop, sum is $0 + 0.1 + 0.2 + 0.3 + \dots + 1.9$

Your answer is correct



Section 5.6 Case Studies

5.19 How many times is the print statement executed?

```
for i in range(10):
    for j in range(10):
        print(i * j)
```

- ☒ A. 100
- ☐ B. 20
- ☐ C. 10
- ☐ D. 45

Your answer is correct



5.20 How many times is the print statement executed?

```
for i in range(10):
    for j in range(i):
        print(i * j)
```

- ☒ A. 100
- ☐ B. 20
- ☐ C. 10
- ☐ D. 45

Your answer A is incorrect



[Click here to show the correct answer](#)

Section 5.7 Keywords break and continue

5.21 Will the following program terminate?

```
balance = 10
```

```
while True:  
    if balance < 9: break  
    balance = balance - 9
```

- ☒ A. Yes
☐ B. No

Your answer is correct



5.22 What is sum after the following loop terminates?

```
sum = 0  
item = 0  
while item < 5:  
    item += 1  
    sum += item  
    if sum > 4: break
```

```
print(sum)
```

- ☐ A. 5
☒ B. 6
☐ C. 7
☐ D. 8

Your answer is correct



5.23 What is sum after the following loop terminates?

```
sum = 0  
item = 0  
while item < 5:  
    item += 1  
    sum += item  
    if sum >= 4: continue
```

```
print(sum)
```

- ☐ A. 15
☒ B. 16
☐ C. 17
☐ D. 18

Your answer B is incorrect



[Click here to show the correct answer](#)

5.24 Will the following program terminate?

```
balance = 10
```

```
while True:  
    if balance < 9: continue  
    balance = balance - 9
```

- ☐ A. Yes
☒ B. No

Your answer is correct



5.26 What will be displayed by after the following loop terminates?

```
number = 25
isPrime = True
for i in range(2, number):
    if number % i == 0:
        isPrime = False
        break

print("i is", i, "isPrime is", isPrime)
```

- ☐ A. i is 5 isPrime is True
- ☒ B. i is 5 isPrime is False
- ☐ C. i is 6 isPrime is True
- ☐ D. i is 6 isPrime is False

Your answer is correct



5.27 What is the number of iterations in the following loop:

```
for i in range(1, n):
    # iteration
```

- ☐ A. 2*n
- ☒ B. n
- ☐ C. n - 1
- ☐ D. n + 1

Your answer B is incorrect



[Click here to show the correct answer](#)

5.29 Suppose the input for number is 9. What will be displayed by the following program?

```
number = eval(input("Enter an integer: "))

isPrime = True
for i in range(2, number):
    if number % i == 0:
        isPrime = False

    print("i is", i)

    if isPrime:
        print(number, "is prime")
        break
    else:
        print(number, "is not prime")
```

- ☐ A. i is 3 followed by 9 is prime
- ☒ B. i is 3 followed by 9 is not prime
- ☐ C. i is 2 followed by 9 is prime
- ☐ D. i is 2 followed by 9 is not prime

Your answer B is incorrect



[Click here to show the correct answer](#)

